

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**

Order Instituting Rulemaking to
Modernize the Electric Grid for a High
Distributed Energy Resources Future.

Rulemaking 21-06-017
(Filed June 24, 2021)

**COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON THE
ADMINISTRATIVE LAW JUDGE'S RULING SEEKING COMMENTS REGARDING
FUTURE GRID STUDY REPORT**

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission (“Commission”), the Vehicle-Grid Integration Council (“VGIC”) hereby submits these comments on the *Administrative Law Judge’s Ruling Seeking Comments Regarding Future Grid Study Report* (“Ruling”), filed October 17, 2024.

I. INTRODUCTION.

VGIC appreciates the opportunity to comment on the Future Grid Study Report (“FGS Report” or “Report”). Central to this discussion is the need to modernize California’s grid to accommodate increasing distributed energy resources (“DERs”) and electrification demands, including for electric vehicles (“EVs”). EVs being added to the California grid is expanding rapidly driven to a large degree by rules and programs from the California Air Resources Board (“CARB”) with the goal to have all new light-duty vehicles sold being zero emissions by 2035. The growth of EV charging load is creating significant challenges for the existing distribution system. Furthermore, California is also experiencing an energy affordability crisis, and while much of the increase in electric utility rates to date has been driven by wildfire mitigation costs, distribution system upgrades are poised to add significant new costs: as high as \$50 billion by 2035 as projected

in the Electrification Impacts Study Part 1, filed in this proceeding on May 9, 2023.¹ In the transition to a high DER future, California should leverage the installed base of customer-sited DERs, including EVs to maximize the use of existing and future distribution infrastructure, while minimizing upgrades to the extent possible. This will be key to ensuring affordability while also meeting the state’s carbon reduction goals.

The Future Grid Study Report provides an overview of earlier working group discussions on the future state of the distribution system with rapidly expanding DER deployments. VGIC believes that the Report provides insightful discussions into many different items, but the Report appropriately acknowledges that there are many overlapping items in scope for both this report, the Smart Inverter Operationalization Working Group (“SIOWG”) Report from Track 3 of this instant proceeding, and work being done in other venues, such as the Demand Flexibility Proceeding (R.22-07-005). With the broad nature of the topics and overlapping scopes in proceedings, VGIC urges the Commission, above all else, to **provide clear venues and explicit next steps for addressing these recommendations, especially for recommendations related to optimizing DER aggregations and enabling flexible load energization.**² In response to the report, VGIC provides the following recommendations:

- A dedicated venue to finalize an effective, scalable flexible load energization framework should be initiated immediately.

¹ *Administrative Law Judges’ Ruling Setting a Workshop, Admitting Into the Record Part 1 of the Electrification Impacts Study and Research Plan, and Seeking Comments*. May 9, 2023. Attachment 2.
<https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M508/K423/508423139.PDF>

² Parties have used various of terms to describe flexible load energization across the instant proceeding and related proceedings. Other common terms or phrases that refer to flexible load energization include: Operational Flexibility in DER Interconnection Agreements, Limited Load Profiles, flexible service connection, and Type II automated load management. Stakeholders have also used flexible load energization to specifically refer to temporary implementation before full-scale energization can be achieved, in which case the terms interim service connection, bridge service connection, scaled service connection, or phased energization are used. In this set of comments, VGIC uses the phrase flexible load energization to remain consistent with the terminology used in the Future Grid Study Report.

- The Commission should focus on “open access to the distribution system” – as defined in the Report – operational needs as a top priority for CA. VGIC recommends a specific near-term focus on the following sub-issues:
 - Improve opportunities for DERs to avoid/defer infrastructure upgrades, including through promoting managed and bidirectional charging programs in addition to dynamic rates adopting least-cost infrastructure planning practices.
 - Improve and simplify interconnection agreements and processes.

II. RESPONSES TO RULING QUESTIONS.

1. The FGS Report includes a wide range of stakeholder input and recommendations from three public workshops. Please review the report (Attachment 1) to ensure it accurately reflects stakeholder input from the workshops. If there are discrepancies, please identify the sections, provide specific details and suggested corrections, and identify any inaccuracies, inconsistencies, or omissions from the workshop discussion in the sections of the FGS Report. Comments should be limited to workshop discussions and proposed recommendations.

VGIC has no comments on this question at this time but reserves the right to comment at a future date.

2. The FGS Report outlines ten key operational needs, categorized as broad themes, that are essential for realizing a High DER future. These needs were identified based on insights and discussion from Workshop 1 - Identifying Operational Needs. (Refer to Attachment 1, pages 24-29, "Outcome: Operational Needs for a High DER Future" section.) Based on the stakeholder survey during workshop facilitation, the FGS Report findings show that the following three operational needs are considered high priority with sufficient urgency to justify implementation within 1-2 years:

- i. DER Visibility to the Distribution System Operator,*
- ii. DER dispatchability/control, and*
- iii. Open access to the distribution system*

a) Do you agree that the above operational needs are the highest priority and need to be implemented within 1-2 years? If so, why?

Sub-issues within “Open Access to the Distribution System” should be prioritized to meet transportation electrification needs while minimizing infrastructure upgrades. Widespread flexible load energization and managed and bidirectional charging programs should be

important priorities for the Commission for implementation in the next 1-2 years.

VGIC agrees that “Open Access to the Distribution System” should place among the highest priority operational needs, but we encourage the Commission to focus on particular distribution optimization items, especially around transportation electrification, to unlock ratepayer benefits in the near term. In the Report, the term “Open Access to the Distribution System” encompasses a wide range of DER integration issues and DER distribution services, including: simplifying interconnection, providing pathways for DERs to reduce distribution investments, integrating DERs into distribution planning, and creating transactive distribution-level markets. This is a very broad definition and VGIC would recommend further breaking down this issue area for implementation and further prioritization of sub-issues.

One of the sub-issues outlined by the Report is to “Meet expected demand for transportation electrification while minimizing infrastructure upgrades.”³ VGIC believes that ensuring that EVs are integrated into the grid in the way that most cost-effectively utilizes our distribution network is one of the most important goals for the California EV transition and this Commission specifically, as it will critically alleviate the state’s compounding affordability crisis. In order to achieve this goal, VGIC recommends focusing on the following “Open Access to the Distribution System” sub-items included in the report as the highest priority for implementation in the next 1-2 years:⁴

- Improve opportunities for DERs to avoid/defer infrastructure upgrades
- Improve and simplify interconnection agreements and process

Creating opportunities for DERs to avoid and defer distribution infrastructure upgrades is key to minimizing the projected costs of widespread electrification. VGIC believes that this should

³ FGS Report at p.26.

⁴ FGS Report at p.26.

be unlocked by allowing the connected load of fleet charging infrastructure to exceed the size of the service connection in the next 1-2 years with widespread Flexible Load Energization (especially for long-dwell, multi-charger sites like workplace charging and fleet depots) and managed and bidirectional charging programs (e.g., targeting residential charging customers). Regarding Flexible Load Energization, the SIOWG Report in Track 3 of this proceeding,⁵ as well as motions by Enphase Energy in this instant proceeding⁶ and R.24-01-018,⁷ have outlined a reasonable framework for discussion. If the Commission were to formally incorporate Flexible Load Energization into an existing proceeding or open a new proceeding with the express intention of finalizing an effective, scalable framework, then Flexible Load Energization options for EV customers could reasonably be implemented at scale in the next 1-2 years.

The Commission can also establish clear direction in the next 1-2 years to reduce the need for infrastructure upgrades by leveraging managed and bidirectional charging programs that are purpose-built to support distribution system optimization. There are multiple dynamic rate pilots that are currently being conducted by the IOUs to explore how to best send price signals to customers that align with real-time grid conditions and marginal costs, including distribution system costs. However, VGIC is concerned the underlying complexity characterizing these rates, designed based on the “CalFUSE” framework, will significantly limit adoption from customers and the automated service providers (“ASP”) needed to unlock customer participation. Additionally, certain detailed elements of the CalFUSE approach, including the subscription

⁵ *Smart Inverter Operationalization (SIO) Working Group Report Business Cases and Use Cases*, February 1, 2024. <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M532/K694/532694061.PDF>

⁶ *Motion Of Enphase Energy, Inc. to Amend Scoping Ruling and Schedule to Address Optional Flexible Connection Agreements and the Underwriters' Laboratory 3141 Standard for Power Control Systems in Investor-Owned Utilities' Energization Rules*, R.21-06-017, May 6, 2024.

⁷ *Motion Of Enphase Energy, Inc. to Amend Scoping Ruling and Schedule to Address Optional Flexible Connection Agreements and the Underwriters' Laboratory 3141 Standard for Power Control Systems in Investor-Owned Utilities' Energization Rules*, R.24-01-018, September 11, 2024.

element which mutes the savings opportunity for customers, may be unattractive to customers that actively seek more exposure to real-time system cost conditions. In the next 1-2 years, these pilots will collect data on customer preference, feasibility of implementation, cost of administration, and, if measurable, impacts on the distribution system costs. To provide a point comparison to the results of these innovative, first-in-the nation dynamic rate pilots – and seeing as it is as yet to be determined the most effective set of strategies to attract the diverse spectrum of EV customers to grid-supportive strategies – the Commission should ensure that non-rate managed and bidirectional charging programs are also widely available. These programs can support meaningful customer engagement in critical, grid-supportive load management and export behaviors. VGIC recognizes that the ongoing affordability crisis makes it challenging to authorize new ratepayer investments, even in programs purpose-built to defer and avoid costly upgrades otherwise borne by ratepayers. However, notably, these same programmatic approaches are identified in the latest amendments to CARB’s Low-Carbon Fuel Standard as example uses of the IOUs’ LCFS Holback Revenues.⁸ **This presents a no-regrets opportunity to establish large-scale programs – *without additional ratepayer funding* – that will help engage customers in managed and bidirectional charging activities not to reduce bulk system-level net peak load, but rather to mitigate secondary peaks⁹ on the distribution system. This is one of the only tools still available to California in its efforts to protect against the projected \$50 billion infrastructure investment expected over the next decade thereby directly addressing the ratepayer burden.**

Importantly, pursuing distribution system optimization through programmatic approaches does not preclude dynamic rate pilot customers from having access to dynamic price signals. The

⁸ *Proposed Amendments to the Low Carbon Fuel Standard Regulation: Attachment A-1: Final Regulation Order*. November 6, 2024. Page 50-51. https://ww2.arb.ca.gov/sites/default/files/barcu/regact/2024/lcfs2024/lcfs_fro_atta-1.pdf

⁹ Alternatively referred to as timer peak or snapback peak.

Report notes that in order to implement broader dynamic distribution pricing, the IOUs/DSOs will have to be able to monitor grid conditions to send accurate price signals to DERs where and when relief is needed. Additionally, the IOUs, CCAs, and other entities will have to utilize MIDAS and other systems to send price signals and create billing systems to handle additional customers on these rates, the latter of which has proven to be a colossal task to date. For these reasons, in addition to the practical limitations on customer and ASP participation identified above, the Commission must act now to ensure programmatic managed and bidirectional charging alternatives to dynamic rate pilots are available at a relatively large scale over the next 1-2 years, including through use of non-ratepayer funds such as LCFS Holdback Revenues. Additionally, distribution system planning must account for the impact of these rates and programs to identify the least cost investments to meet EV charging needs.

The Commission should also prioritize streamlining Rule 21 generator interconnection for vehicle-to-everything (“V2X”) bidirectional charging resources. V2X bidirectional charging systems have significant potential to provide both local and systemwide grid benefits, but, to unlock this, key interconnection barriers need to be removed. There is a Rule 21 interconnection process for bidirectional chargers, but several key barriers remain. For example, high interconnection application fees (i.e., \$800¹⁰ - the highest in the nation) are imposed on these customers. Additionally, not all IOUs have appropriate interconnection application forms for V2X bidirectional charging systems, resulting in applications being delayed or risking being knocked from the generator interconnection process into the load energization process. These items do

¹⁰ See respective Rule 21 pages:

<https://www.pge.com/en/about/doing-business-with-pge/interconnections.html?ctx=business>

<https://www.sce.com/business/generating-your-own-power/Grid-Interconnections/Interconnecting-Generation-under-Rule-21>

<https://www.sdge.com/more-information/customer-generation/electric-rule-21>

require discussion, likely in a forthcoming Rule 21 proceeding, but it is reasonable – and necessary to maximize cost-effective VGI – to achieve significant progress in resolving these issues in a 1-2 year timeframe.

In contrast, many of the other “Open Access to the Distribution System” sub-issues around transactive markets are likely unable to be implemented in the near-term. VGIC is supportive of further discussions and developing non-discriminatory procedures for procuring, dispatching & compensating DERs for distribution grid services. However, there are significant barriers to overcome to create a centralized distribution-level market to receive bids for different services and run a clearinghouse like the CAISO currently does for the wholesale energy market. The Report discusses the need to create a distribution data sharing hub to allow for sufficient data sharing to facilitate trading and scheduling of DERs.¹¹ VGIC supports this initiative, and notes there are other proceedings addressing some of these sub-issues, such as the Data Working Group in R.23-12-011. Given the complexity of a border transactive distribution services market, VGIC recommends continuing discussions on the topic but focusing Commission resources on the near-term, low-cost to ratepayers, and no-regrets actions described above to enable widespread EV charging while mitigating distribution system investments. This will allow stakeholders to focus on the highest priority, most urgent items that can realistically be implemented in a 1-2 year timeframe.

With this in mind, VGIC recommends that the Commission focus on implementation of the specific sub-items needed to unlock flexible load energization, non-rate managed and bidirectional charging programs, and streamlined interconnection. Full DER visibility, for example, is not likely needed to enable these near-term items and is also unlikely to be

¹¹ FGS Report at p.27.

implemented in the near-term. VGIC discusses further issues of visibility and control of DERs in response to question 3.

b) If you disagree with any of the above-identified high-priority needs, which three operational needs should be prioritized and why? Please include a timeline for implementation and explain your reasoning for supporting the priorities.

VGIC has no additional comments on high-priority needs at this time, but reserves the right to comment at a future date.

3. Diverging approaches to enabling a High DER Future— Following Workshop 1, the workshop series highlighted a diverging approach to long-term visions for a High DER Future between the IOUs' top-down "grid orchestration" approach where DSOs are central in coordinating DERs and the bottom-up, open-access vision recommended by other stakeholders. (Refer to Attachment 1, page 36, "Key Takeaways from Workshop 2".)

a) Which approach do you support, the top-down "grid orchestration" approach or the bottom-up, open-access vision for a High DER future? Please explain your reasoning for supporting the chosen approach and what steps should be taken in this proceeding to implement this vision. How do these steps align with the DSO's roles and responsibilities?

VGIC does not agree that these approaches are mutually exclusive and, as such, supports elements of both visions. For lack of a better term, VGIC believes a "hybrid approach" will best support affordability and serve the state's decarbonization goals. Open access to the distribution grid and customer-centric market design must be fundamental principles of the High DER future, however, there will remain an essential role for the DSO to play in monitoring, interpreting, and communicating real-time grid conditions, including to qualified ASPs and other third parties that will enable the effective integration of DERs onto the grid.

Ensuring that DERs can connect to the grid and begin providing services will help California maximize the value of these customer resources, especially EVs. California is in a unique position as the leading EV market in the United States, having recently achieved the milestone of selling 2 million EVs, but being a leader means that the best path forward to unlocking

value from these vehicles is less clear. Aiming for an open access vision will allow California to move quickly to unlock value and maintain flexibility when it comes to creating value from these resources. For example, if it is easy for customers to *connect* EV load (or, in the case of V2X bidirectional charging systems, *interconnect* as energy storage) to the distribution grid and enroll in a variety of efforts, they can choose whether to shape load under a dynamic rate tariff, participate in more traditional demand response or managed and bidirectional charging programs via a utility, CCA, or third-party, or, in the case of V2X bidirectional charging systems, discharge to meet site load. Given the immense scale of EV load flexibility and export capability anticipated over the next decade, the huge diversity of defined VGI use cases (e.g., the over 1,000 distinct VGI use cases detailed extensively in the 2019-2020 Joint Agency VGI Working Group¹²), the growing affordability crisis and its high stakes, and the underwhelming portfolio of VGI opportunities to date in California, it would be imprudent for the Commission, VGIC, or any other party to foreclose on any of these options at this time.

If the DSO plays a very active “orchestration role,” where all DERs are enrolled with the DSO/IOU and even controlled by the DSO, exploration options will be limited. Having all devices enrolled with a DSO would likely limit offering and not as many customers would find program options that work for them. On the other hand, the DSO does hold unique visibility into the distribution system as a whole and specific expected and real-time grid conditions. As a result, VGIC envisions an important role for the DSO providing data for DERs to respond to. These signals can be **price/market-based**, where the DSO is communicating a real-time price to buy or sell energy, **event-based**, where the DSO is signaling a demand response event that customers

¹² California Public Utilities Commission. Final Report of the California Joint Agencies Vehicle-Grid Integration Working Group. June 30, 2020. <https://gridworks.org/wp-content/uploads/2020/07/VGI-Working-Group-Final-Report-6.30.20.pdf>. Page 15.

may be obligated or encouraged to respond to, or **communicate changes to physical import or export limits and settings**. Some elements of Flexible Load Energization and Limited Load Profiles, as well as dynamic Limited Generation Profiles, might include more active “grid orchestration,” where the DSO can send signals to power control systems that respond to limit imports or exports in real-time. This example shows that some level of grid orchestration and more explicit direction from the DSO will likely be a part of the High DER future.

b) How do the operational needs identified in question 2 above align with your recommendation for question 3 (a) above?

VGIC’s recommended priority operational needs discussed in question 2, particularly the recommendations to create a flexible load energization pathway and establish programmatic managed and bidirectional charging approaches, are focused on unlocking the most practical and near-term value from DER orchestration while maintaining customer optionality. Currently, VGIC can envision a future composed of more complex elements such as transactions between customers for additional import / export capacity. However, those complex market structures will take significant amounts of time to develop. In the near-term, the DSO can continue to play an important role in supporting DER optimization to unlock significant ratepayer benefits, both through dynamic rate implementation and programmatic approaches, especially those supported by non-ratepayer funds.

c) Based on the discussions and perspectives presented during the workshop series, do stakeholders envision a hybrid approach that could bridge the gap between the IOU vision of grid orchestration and the bottom-up, open-access vision? If so, what might such a hybrid model entail?

As discussed above, VGIC does envision a hybrid approach for the High DER Future. The Commission should seek to keep the DER service market open to allow additional DERs to provide

grid services. However, a strong DSO role is appropriate, and customers should receive compensation commensurate with the full stack of services they are providing.

4. In Workshop 2, Assessing Gaps, the stakeholders focused on identifying gaps and barriers to achieving the operational needs from Workshop 1. During Workshop 2, the IOUs also presented roadmaps for DERs and/or Grid Orchestration and Automated Distribution Management Systems (ADMS) and DER Management Systems (DERMS) capabilities in relation to the operational needs identified from Workshop-1. (See pages 58-82 of Attachment 1, "APPENDIX A: OPERATIONAL NEEDS GAP ASSESSMENT" for a detailed assessment of the gaps.) The following questions focus on operational needs and use cases articulated specifically for DER/Grid Orchestration by leveraging ADMS/DERMS capabilities.

a) Which operational needs and use cases can be operationalized by addressing gaps and barriers to DER/Grid Orchestration and ADMS/DERMS capabilities? How do these steps align with the DSO's roles and responsibilities?

Improvements to ADMS/DERMS are needed to unlock widespread flexible load energization for flexible import capacity that changes dynamically over time. These changes to import limits would have to be communicated by the DSO to a customer site with a flexible energization agreement. PG&E states that they have “plans to scale DERMS capabilities [...] to communicate load limits day ahead [...] and are] looking to scale across territory.”¹³ Ultimately, it is the DSO that will have visibility into whether there is additional capacity on a given circuit and can provide signals that allow customers to take advantage of times when additional capacity is available on the circuit.

b) What specific steps should be addressed in this proceeding to advance DER operationalization? How do these steps align with the DSO's roles and responsibilities?

The first and immediate steps that should be taken by the Commission is to enable broad-scale flexible load energization and managed and bidirectional charging programs to unlock latent EV grid-support capabilities. VGIC agrees with the recommendation by Cal

¹³ FGS Report at p.60.

Advocates that “The CPUC should lead a process to develop and implement flexible import limits, following-on the work of the SIOWG.”¹⁴ The DSOs, currently the IOUs, are responsible for energizing new load on the distribution system. The Commission has the authority to direct the development of these programs, but it will require active steps to be taken by the IOUs in order to ensure that rules work with their current DSO processes. This work will likely span proceedings, which VGIC discusses further below in response to question 5e.

c) The IOUs recommended a working group/task force(s) to collaborate on a framework that enables advanced DER/Grid Orchestration. Do you agree with this recommendation, and if so, what are the key factors to consider in forming and setting goals for work products for this group? If you disagree, what alternative approaches should be considered?

VGIC does not offer feedback on a working group on DER/Grid Orchestration at this time but may offer replies to party comments.

d) In Workshops 1 and 2, the IOUs outlined a “grid orchestration” vision for a high-DER future. The terminology and Grid Modernization Plans (See Attachment 1, Appendix B) indicate a significant role for DSOs in orchestrating DERs. However, despite pilot work underway (See Attachment 1, Appendix C), there are still gaps in the timelines for developing and implementing operational capabilities. What improvements should be made to the ongoing pilot program? How can the ongoing and proposed pilots guide further actions to address the gaps? In your response, identify the gaps and the proposed enhancements.

Currently, both PG&E and SCE are conducting flexible load energization pilots, specifically for customers who cannot fully energize their sites on a 24/7 basis at this time because of triggered distribution upgrades. VGIC does not have recommendations to change the pilots at this time, but we believe that these pilots will provide important insights for the broader discussion on a flexible load energization framework and rule modifications. We agree with Gridworks that

¹⁴ FGS Report at p.51.

the results of these pilots should be shared, ideally in a venue focused on finalizing a framework that would facilitate the evolution from individual pilots to full-scale offerings. VGIC respectfully urges the Commission to clearly identify this venue.

e) Is there a benefit in reviewing the five nearest-term pilot programs or functions the utilities can roll out and determining the technical requirements for those options? Explain your reasoning.

In the Report, the California Solar & Storage Association made this recommendation for the Commission “to look at the five nearest term programs or functions that the utilities would like to roll out and determine the technical requirement for those options.”¹⁵ VGIC supports the intention and practicality of this recommendation, but would recommend focusing on the five programs/functions the Commission is prioritizing, not solely what the utilities would like to implement. The Commission is in the best position to consider input from all stakeholders to determine which items should be prioritized for implementation.

5. The FGS Report compiled stakeholder recommendations as five topic summaries based on input and discussion during Workshop 3 - Developing Recommendations to Address Gaps. The FGS Report also includes Gridworks’ recommendations for the next steps. (Refer to the “Stakeholder Recommendations ...” and “Gridworks Recommended Next Steps ...” sections in Workshop 3, pages 39 – 51 of Attachment 1.)

e) If any of your recommendations require coordination with other proceedings beyond the scope of the High DER Proceeding, please provide specific details about the necessary coordination, including the relevant proceedings and issues involved.

For flexible load energization, coordination is needed with the Energization Timelines Rulemaking (R.24-01-018) in order to update IOU energization rules to include flexible energization. As discussed by the SIOWG Report, the full implementation of Flexible Load

¹⁵ FGS Report at p.51.

Energization will require updates to Rule 2, Rule 15, Rule 16, as well as Rules 29 and 45 for EVs.¹⁶ The Commission has already stated that “Issues related to energization rules should be addressed in the energization proceeding.”¹⁷ At the same time, the Commission has issues discussed in the SIOGW that are in scope for this instant proceeding, including: technical operational details of power control systems, structure and temporal granularity of limited load profiles or schedules, and methods of allocating available Firm and Non-Firm distribution system hosting capacity. The Commission has not ruled on Enphase's motion on Flexible Load Energization in R.24-01-018, so the exact split of issues between the two proceedings remains unclear. Regardless, cross-proceeding coordination will be critical.

To further non-rate managed and bidirectional charging programs for distribution system optimization, coordination is needed with the Transportation Electrification Proceeding, R.23-12-008. While this instant proceeding may discuss how to unlock distribution services and DER optimization, EV program funding issues, including authorization of LCFS Holdback-funded programs, is in scope in R.23-12-008.

Streamlining interconnection for V2X resources as discussed in response to question 2a will require coordination with a Rule 21-focused proceeding. The current Rule 21 Rulemaking, R.17-07-007, is planned to close, but the Commission has indicated the intention to “open a new proceeding that focuses on issues affecting interconnection of distributed energy resources under Electric Rule 21.”¹⁸ Once that proceeding is opened, the Commission should include several issues in the proceeding scope that can meaningfully support V2X bidirectional charging systems,

¹⁶ *Smart Inverter Operationalization (SIO) Working Group Report Business Cases and Use Cases*, February 1, 2024, at p. 115. <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M532/K694/532694061.PDF>

¹⁷ *Administrative Law Judges' Ruling Denying Enphase Motion*, June 21, 2024, at p. 1. <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M534/K105/534105698.PDF>

¹⁸ *Proposed Decision Closing Proceeding*, R.17-07-007, November 15, 2024. <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M546/K381/546381906.PDF>

including reducing the \$800 interconnection application fee and directing basic but critical functionality be incorporated into each IOUs' interconnection application forms to apply appropriate treatment to V2X bidirectional charging systems.

III. CONCLUSION.

VGIC appreciates the opportunity to provide these comments on the Report. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen

Zach Woogen

Interim Executive Director

VEHICLE-GRID INTEGRATION COUNCIL

Date: December 6, 2024